ENG M 401 Formula Sheet

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Interest Rates

1. **Bank of Canada Overnight Rate**: The rate set by the central bank, most referenced in media
2. **Prime Interest Rate**: The rate banks set for creditworthy customers, variable mortgages ($\geq$ Overnight Rate)
3. **10 Year Treasury Yield (Risk-Free Rate)**: The rate you can get in market risk-free, followed by fixed mortgages
4. **Inflation Rate**: Measured by Consumer Price Index (CPI). Percent increase of 'basket of goods'. Effect of monetary policy.

Interest, Compounding, Doubling Time

Simp. $\implies F = (1 + i \times n)P$ Comp. $\implies F = (1 + i)^n P$

$$i_r = \frac{1 + i_n}{1 + i_i} - 1 \qquad n_{dbl} = \frac{ln(2)}{ln(1 + i)} \approx \frac{72}{100 \times i}$$

$$EAIR = \left(1 + \frac{i}{m}\right)^m - 1 \qquad \text{Cont. Cmp.} \rightarrow EAIR = e^i - 1$$

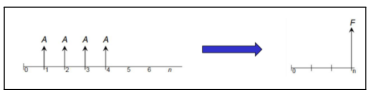
Time Value of Money

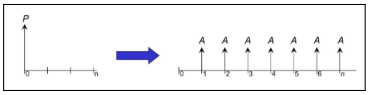
Depends on inflation, risk, liquidity, supply/demand.

Perpetuity

$$P = \frac{C}{i}$$

Future/Present Value of Uniform Series

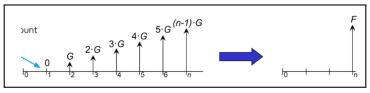


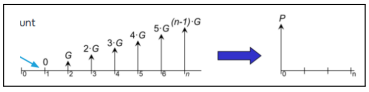


$$F = A \times \frac{(1 + i)^n - 1}{i} \qquad A = F \times \frac{i}{(1 + i)^n - 1}$$

$$P = A \times \frac{(1 + i)^n - 1}{i \times (1 + i)^n} \qquad A = P \times \frac{i \times (1 + i)^n}{(1 + i)^n - 1}$$

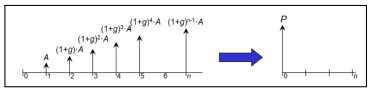
Present/Future Value of Arithmetic Gradient





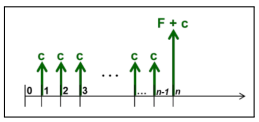
$$F = G\left(\frac{(1 + i)^n - i \times n - 1}{i^2}\right) \qquad P = G\left(\frac{(1 + i)^n - i \times n - 1}{i^2 \times (1 + i)^n}\right)$$

Present Value/Geometric Gradient



$$P = A \times \left(\frac{1 - (1 + g)^n \times (1 + i)^{-n}}{i - g}\right) \xrightarrow{i=g} P = \frac{A \times n}{1 + i}$$

Bond Pricing



1. Calculate coupon rate $c = Par \times (Coupon\ Rate/2)$ (i.e., use semi-annual value for gov't bonds)
2. Calculate yield to maturity (use semi-annual for gov't bonds)
3. Discount bond cashflows including par value to present value using $i = YTM/2$, $c = coupon$, and $F = Par$.
4. Disc. bond $\implies$ PV $\leq$ par, prem. bond $\implies$ PV $\geq$ par

Note: If interest rates rise, bond price falls and if interest rates fall, bond prices rise

Mortgages

- **Amortization** Time in years mortgage will take to pay off.
- **Term** Length of contract between borrower and lender (usually 5, 1-10yrs). Pay at **Payment Frequency**
- **Equity** Current value of home - remaining principal
- **Fixed/Variable Rate** Fixed rate has set rate. Variable/Adjustable payments fluctuate to keep interest/pricipal paid the same, Variable/Fixed payments stay same, share of interest/principal varies. Use (A/P, i, n) to find payments.

Equipment Cost

- **Ownership Costs**: Fixed costs incurred each year. Includes depreciation, investment, insurance, tax, storage
- **Operating Costs**: Variable costs incurred each year. Includes FOG/consumables, repair/maintenance, tires/high-wear, operator salary

Equipment Cost Steps

1. Separate/deduct cost of tires/tracks from total cost
2. Calc. PV of depreciation and other ownership costs, add/subtract from equip. cost, spread as annuity, calculate hrly cost
3. Calc. lifetimes of tires/wear items, how often they must be replaced, and PV of replacement cost. Spread as annuity, calculate hrly cost

4. Calculate NORMAL SL dep of items that need repair (n = ITEM LIFETIME), multiply by factor, convert to hrly
5. Calculate hrly FOG/consumable/operator costs, add to repair/replacement costs to find operating cost

Rent, Buy, Lease, Replace

$\implies$ **Rental** best for short-term/testing/atypical needs. **Lease** good for managing risk/avoiding upfront cost. **Buy** best for long-term/frequent use

$\implies$ Replace equipment when maximum benefit achieved, i.e. ownership/operating cost total at min.

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Present Value Analysis

- **Nominal Dollars**: The money we normally think of, their value is reduced due to inflation
- **Read Dollars**: Also called inflation-free dollars. Keep their value.

Net Present Value Analysis (NPV)

Discount all cashflows to present day, perform project if NPV ≥ 0. Use incremental cash flow defined on incremental basis, remember working capital, consistent inflation, consider investment and financing decisions separately, trust market prices.

Capitalized Cost Analysis

1. Draw cash flow for two cycles, find PV of non-recurring cash flows
2. Find uniform annual series for one cycle, find $PV = C/r$
3. Find total present value

Competitive Bid Process (RFQ/RFP)

Steps: Preparation/Solicitation $\implies$ Bid Submission $\implies$ Bid Evaluation $\implies$ Selection and Award $\implies$ Execution.

Key source of risk is in preparing technical specifications $\implies$ use strategic pricing to take advantage of changes in scope/assumptions

Managing Risk

Risk can be managed in three main ways: **Contingencies (blind allowance, reflect stage of project), Increase Required Returns/Discount rate, Mitigation** (ex. risk matrix)

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Rates of Return

$WACC = W_D k_D (1-T) + W_E k_E$ MARR $\geq$ WACC

- **Payback Period:** Amount of time it takes for net benefit to equal net cost. **Simple** $\rightarrow$ no discounting, **Discounted** $\rightarrow$ use discounting. Only use as tiebreaker/projects with high uncertainty
- **Internal Rate of Return (IRR):** Discount rate at which NPV=0. Perform project iff IRR$\geq$MARR. Find **Incremental IRR** by taking cash flows of cheaper project, subtracting from more expensive, find if ΔIRR$\geq$ MARR.
- **Internal Rate of Return (IRR):** Test if $PW_{benefits}/PW_{costs} \geq 1$. Note: $BCR > 1 \implies NPV > 0$. Use for public sector projects. To compare, calc. **Incremental BCR** and test $\Delta BCR \geq 1$
- **Sensitivity/Breakeven Analysis:** Calculate what values of specific parameters lead to breakeven scenario (NPV=0)

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Market Valuation

Income-Based Methods

Best for businesses that generate profit/those whose value exceeds sum of assets

1. **Discounted Cash Flows:** Calculate DCFs of business
2. **Capitalization of Earnings:** Value=Net Income or earnings divided by discount rate or capitalization rate
3. **Income Multiplier:** Multiply EBIT or EBITDA by factor. (ex. 18x for banks, 12x advert./marketing, 18x aerospace, 10x utilites, 8x homebuilding)
4. **Gordons's Dividend Discount Model:** Calculate $PV_0 = D_1/(r-g) = D_0(1+g)/(r-g)$. D_1 is next dividend PER share. Multiply by number of shares outstanding to find business value.

Market-Based Methods

Best for private businesses and those without strong history of revenue/profit

1. **Market Capitalization:** Calc. share price times number of shares outstanding
2. **Comparable Company Analysis (CCA):** Use key financial metrics of company and compare to publicly traded ones. Use for private companies
3. **Precedent Transaction Analysis (PTA):** Assess valuation multiples (ex. of EBIT) by acquirers of similar companies

Asset-Based Methods

Best for asset-heavy businesses or when business is to be liquidated. Typically gives lowest valuations

1. **Book Value:** Calc. Total Assets - Total Liabilites. Not the best method.
2. **Liquidation Value:** Calculate business value if assets sold off and liabilities settled. Typically less than Book Value. Use for companies that face distress/bankruptcy
3. **Replacement Cost:** Value company as cost of replacing assets at current market prices